UE22CS441A – Capstone Project Phase – 3

Project Progress Review # 1

Cognitive Load Estimation from EEG + fNIRS Signals using Hybrid Deep Learning

Project Title : Cognitive Load Estimation from EEG + fNIRS

Project ID :

Project Guide :

Project Team :

Outline

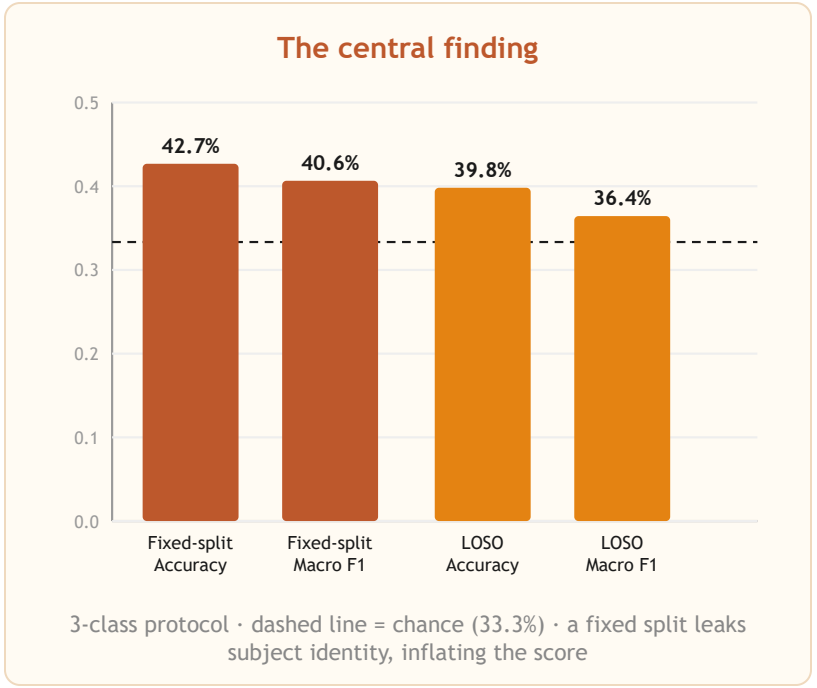
- Abstract and Scope of the Project
- Summary of Work Done in Capstone Project Phase – 2
  - Preprocessing, hybrid model, and leakage-audited LOSO evaluation
  - Binary reframing and the 42-experiment improvement campaign
- Inferences Drawn from Literature Survey
- List of Tasks / Modules with Individual Contribution
- Design of Overall Architecture
- Demonstration and Testing of Completed Modules
- Key Results — Confusion Matrix & Per-Subject Performance
- Explainability — SHAP & Attention Analysis
- Experiment Campaign — Rigor & Negative Results
- References

Abstract and Scope

Mental workload cannot be measured directly — it has to be inferred from physiological signals. This project estimates cognitive load from two complementary brain signals recorded simultaneously: **EEG** (electrical activity, 200 Hz, fast but noisy) and **fNIRS** (hemodynamic/blood-oxygenation activity, 10 Hz, slower but stable), using the open **TU Berlin n-back dataset** (Shin et al., 2018) — 26 subjects performing 0-back / 2-back / 3-back working-memory tasks, an experimentally imposed workload level that gives objective ground truth.

A **hybrid deep-learning network** — per-modality residual-MLP encoders fused through a learned attention gate — predicts Low / Medium / High load (original protocol) or Low / High (binary reframing).

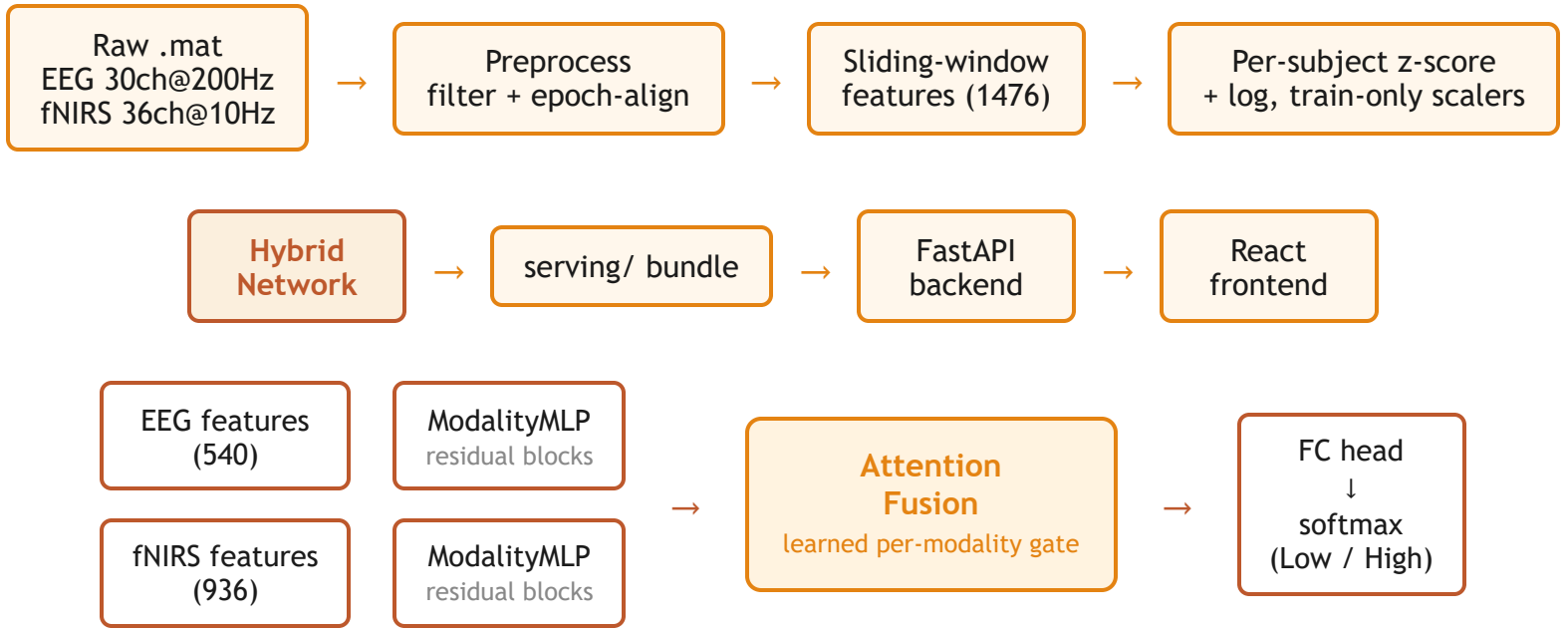
Scope: (1) an end-to-end, leakage-audited pipeline; (2) an honest headline metric — leave-one-subject-out (LOSO) accuracy, not an inflated fixed-split number (right); (3) a full-stack demo where every UI number traces to a committed artifact; (4) SHAP + attention-gate explainability; (5) a documented negative result (DANN) as evidence of rigor, not hidden.



Summary of Work Done in Capstone Project Phase – 2

- Built the full preprocessing pipeline: raw `.mat` recordings → band-power / hemodynamic features → sliding-window extraction, with EEG epoched on NIRS markers to keep both modalities aligned.
- Trained and evaluated the 3-class Hybrid network; identified the **fixed-split-vs-LOSO gap** as the project's central methodological finding.
- Adopted per-subject z-score + log-compression preprocessing; ran a 26-fold LOSO rigor pass with paired significance tests and Random-Forest ensembling.
- Explored a **Domain-Adversarial Neural Network (DANN)** for subject-invariant features — documented as a rigorous, honest negative result.
- Reframed the task as **Binary (Low vs High)** with sliding-window augmentation, lifting LOSO accuracy from ~40% to **63.09%**.
- Built a leakage-audited experiment framework (`research/`) and ran **42 controlled experiments** on the binary model against a development/lockbox split — none beat the baseline within one standard error.
- Exported the frozen model to a portable `serving/` bundle and built the FastAPI backend + React frontend demo, including a CSV-upload endpoint verified byte-exact against the training pipeline.

Architecture



The attention gate is exposed in every API response — it is what the app's modality-contribution bar shows (how much the model trusts EEG vs. fNIRS for that prediction).

List of Tasks / Modules

- **Data Preparation** — `preprocess_tuberlin.py`, `sliding_window_extractor.py`, `data_loader.py`: raw `.mat` → filtered, epoch-aligned, per-subject-normalized feature tensors (1476 features/window).
- **Model Architecture & Training** — `models/architectures.py`, `trainer.py`: attention-fusion Hybrid network; validation-macro-F1 checkpoint selection, class weighting, early stopping.
- **Rigor / Research Framework** — `research/` (`leakage.py`, `datamod.py`, `train.py`, `campaign.py`): leakage-audited LOSO folds, the 42-experiment binary-model campaign, development/lockbox protocol.
- **Explainability** — `explain_model.py`, `visualize_attention.py`: SHAP (GradientExplainer) + attention-gate analysis.
- **Export & Packaging** — `export_for_serving.py`, `export_research_data.py`: freeze model + artifacts into portable, versioned bundles.
- **Backend API** — FastAPI (`main.py`, `model_service.py`, `data_service.py`, `upload_service.py`): inference, CSV-upload validation, run-archive access.
- **Frontend UI** — React + Vite, 7 pages: Overview, Research, Live Prediction, Session Trend, Upload Data, Model Insights, Methodology.

Tools / technologies (all open-source): Python 3.14, PyTorch, scikit-learn, SHAP, pandas / NumPy, FastAPI, Pydantic, React 18, Vite, Recharts.

Individual Contribution

Module	Approx. Lines of Code	Description	Owner
Data & Training Pipeline	~10,600	Preprocessing, model training, evaluation, ablation scripts (top-level *.py)	
Research / Leakage-Audit Framework	~5,070	Binary-model campaign, leakage assertions, run archive (`research/`)	
Backend (FastAPI)	~1,630	Serving API, model inference, CSV upload (`backend/`)	
Frontend (React / Vite)	~9,330	7-page demo UI, charts, explainability views (`frontend/src`)	
Total	**~26,600 lines**	Across 202 tracked files, Python + JS/JSX	

Single-member team — all modules designed, implemented, evaluated and documented by the contributor named above. Timeline: Phase 1 – dataset, preprocessing, 3-class model; Phase 2 – leakage-audited framework, binary reframing, backend/frontend demo; Phase 3 (current) – frontend redesign, reporting.

Demonstration and Testing of Completed Modules

Protocol (Binary model)	Accuracy	Macro F1	ROC-AUC
LOSO, 26 subjects (4,576 pooled windows)	**63.09% ± 10.1%**	0.6255	0.6775
Lockbox, 8 unseen subjects (evaluated once)	**63.71% ± 8.5%**	0.6275	0.6888

- **Live Prediction demo** — pick a subject + window, run a real forward pass; the app shows predicted class, probability bars, confidence / margin / entropy, and the attention-gate bar (how much the model trusts EEG vs. fNIRS).
- **Automated correctness check** — `research.verify_run` recomputes every metric from each run's saved `predictions.csv` and reports zero mismatches.
- **Leakage testing** — subject / trial / window / duplicate-row / scaler-fit / held-out-subject assertions run before every fold trains; a failure raises `LeakageError` and aborts the run.
- **Upload-path testing** — the CSV-upload preprocessing is a self-contained reimplementation of the training pipeline, verified byte-exact (max abs diff 0.0) against `data_loader.py`.

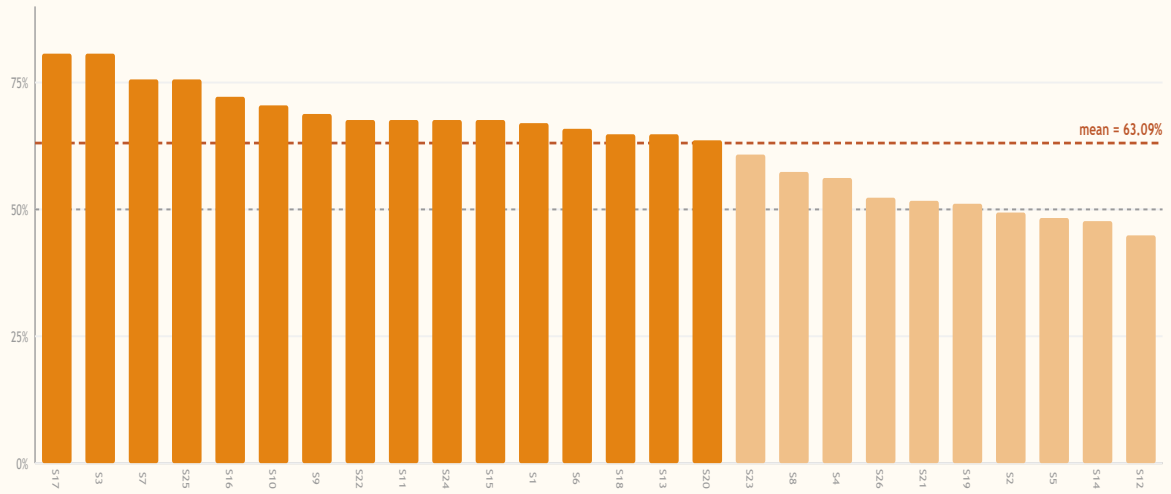
Key Results — Confusion Matrix & Per-Subject Performance

Binary model — pooled LOSO confusion matrix

	Pred. Low	Pred. High
True Low	1561 68.2% recall	727
True High	962	1326 57.9% recall

4,576 pooled held-out windows, 26-fold LOSO · overall accuracy 63.09%

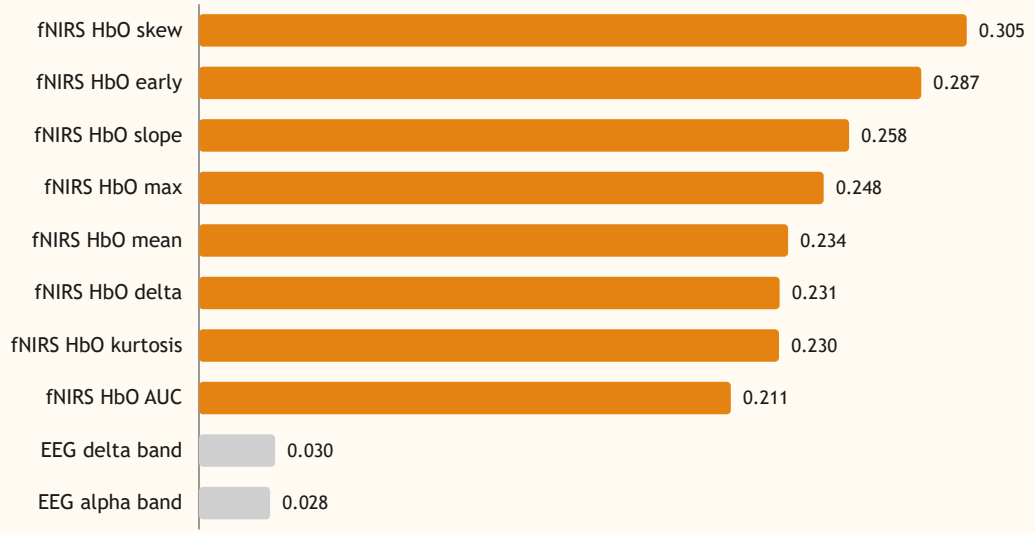
Per-subject held-out accuracy (26 subjects)



Best: S17/S3 at 80.7% · worst: S12 at 44.9% — the spread across people is wider than any modelling change tried

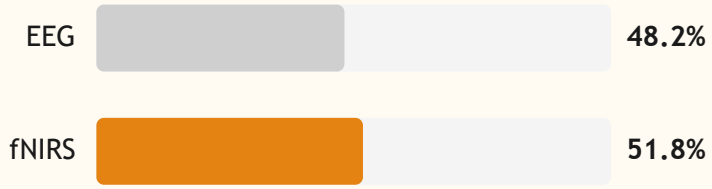
Explainability — What Drives the Prediction

SHAP — top feature categories (served model)



fNIRS HbO response-*shape* features (skew, early-window, slope) dominate; EEG bands contribute an order of magnitude less

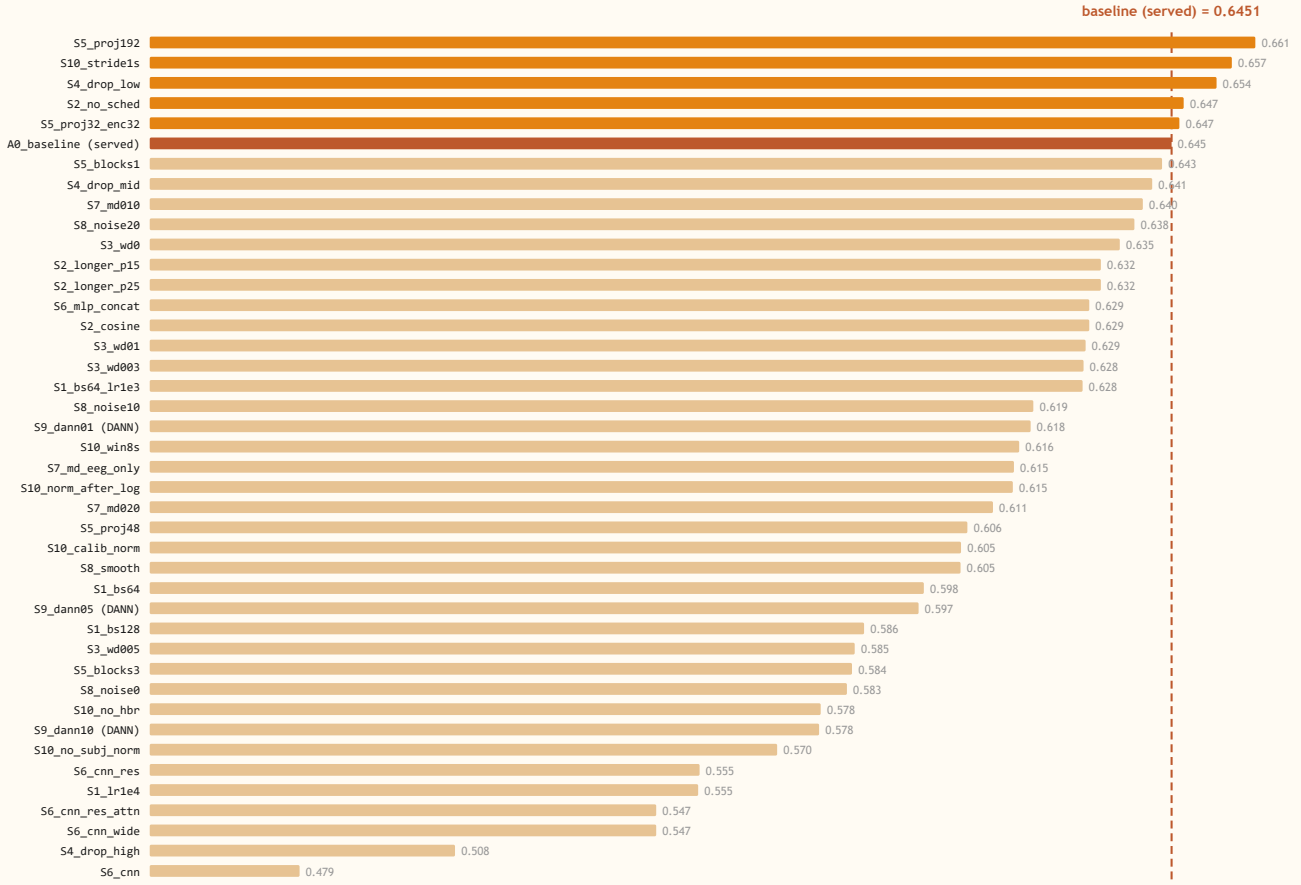
Learned attention gate (mean, all classes)



Two independent methods (SHAP, attention) agree: the model leans on fNIRS but does not ignore EEG

- SHAP (`GradientExplainer`) and the network's own learned attention gate were computed independently — both point the same way, which is the reassuring part.
- The attention gate is exposed in every live API response; it is what the app's modality-contribution bar shows for each individual prediction.

Experiment Campaign — Rigor & Negative Results



Orange = at/above baseline · tan = below baseline · burnt-orange = served baseline (A0). None of the 42 variants (DANN, architecture, capacity, augmentation, normalization axes) cleared the one-standard-error acceptance bar, so the plain Hybrid network is what's served — a documented negative result, not a gap to hide.

References

1. J. Shin, A. von Lühmann, B. Blankertz, D.-W. Kim, J. Jang, H.-J. Hwang, and K.-R. Müller, “Simultaneous acquisition of EEG and NIRS during cognitive tasks for an open access dataset,” *Scientific Data*, vol. 5, no. 180003, 2018.

2. Y. Ganin and V. Lempitsky, “Unsupervised Domain Adaptation by Backpropagation,” in *Proc. 32nd Int. Conf. on Machine Learning (ICML)*, 2015, pp. 1180–1189.

3. S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.

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5. R. M. Yerkes and J. D. Dodson, “The relation of strength of stimulus to rapidity of habit-formation,” *J. Comparative Neurology and Psychology*, vol. 18, no. 5, pp. 459–482, 1908.

Thank You